Run report: cen-q5-split-run-20261024T2206Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (sampled) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: cen-q5-split 20261024T2206Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 cc0bcbe7b428feff4e2acbea3c6887d6abb5031b01f3a49589bd3597b3773dc7
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 457dc80ceded9acbbb0fb39ca1c61e8c3e2bfa773f863100584969c76131187d
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/cen-q5-split-run-20261024T2206Z CENSUS 5 split 50000000 202610245 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 108
```

tool_log.txt

```
split q=5: samples=50000000 (random) res_shaped=82699 allowed_trace_candidates=2476 smooth_prime_order_hits=2476 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1109/2000 secs=107.8
split q=5: total prime-order mass estimate 65.592 +- 1.318 (hit rate per sampled form 4.952e-5)
```

summary.txt

```
split q=5: samples=50000000 (random) res_shaped=82699 allowed_trace_candidates=2476 smooth_prime_order_hits=2476 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=1109/2000 secs=107.8
split q=5: total prime-order mass estimate 65.592 +- 1.318 (hit rate per sampled form 4.952e-5)
```

cand_t_hist.csv (first 12 of 90 rows)

type	t	candidates
split	−49	133
split	−48	86
split	−47	28
split	−46	132
split	−45	110
split	−44	79
split	−43	79
split	−42	99
split	−41	151
split	−40	707
split	−39	449
split	−38	119

hits.csv (first 12 of 2476 rows)

type	code	t	nE	smooth
split	34174412965	39	587	true
split	41046789191	7	619	true
split	4372281092	9	617	true
split	56543983617	25	601	true

split	6772917693	19	607	true
split	72035448669	9	617	true
split	46098340638	7	619	true
split	102118621842	−27	653	true
split	120158365015	27	599	true
split	91157232527	39	587	true
split	102429001849	19	607	true
split	133188596408	9	617	true

mass_by_t.csv (first 12 of 12 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
split	−47	673	28	0.7417	0.1402
split	−33	659	179	4.7419	0.3544
split	−27	653	319	8.4506	0.4731
split	−21	647	170	4.5035	0.3454
split	−17	643	148	3.9207	0.3223
split	7	619	191	5.0598	0.3661
split	9	617	394	10.4374	0.5258
split	13	613	180	4.7684	0.3554
split	19	607	176	4.6624	0.3514
split	25	601	432	11.4441	0.5506
split	27	599	183	4.8478	0.3584
split	39	587	76	2.0133	0.2309

Graph

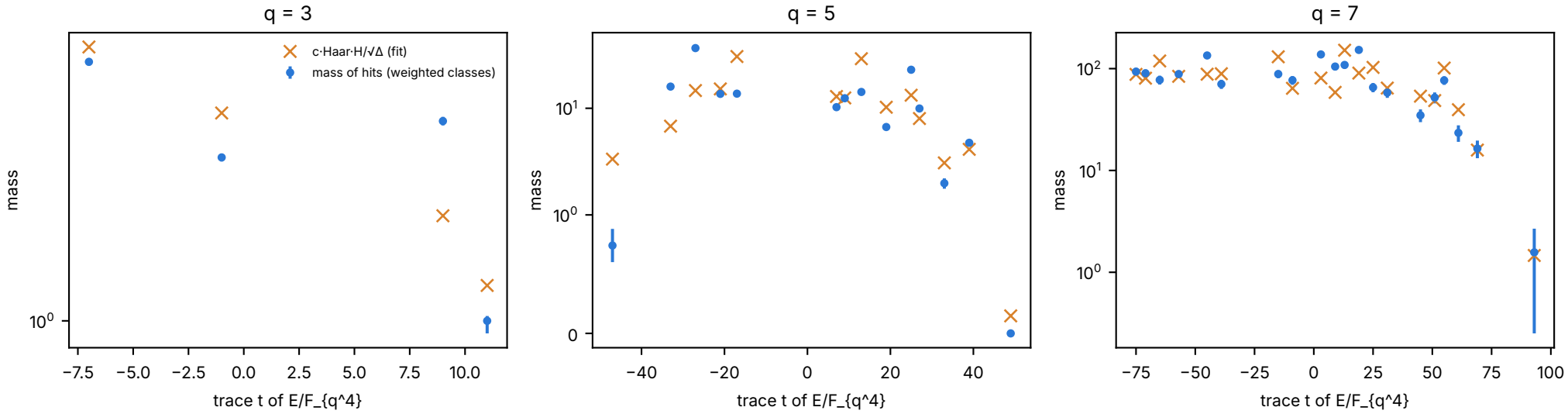


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.